

Unit 5

Post-SPE linear models

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LNG 201 Phonology

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Winter 2022

Readings

EPG Pathashala Paper 5, Module 5 – Abstractness controversy II

Clark & Yallop (1995) – Pg. 402 to 405

Katamba (1989) – Pg. 109 to 115

Lass (1984) – Pg. 223 to 228

Seminal pieces

- Vennemann, Theo (1972). Phonological uniqueness in natural generative grammar. *Glossa* 6: 105-61.
- **Vennemann, Theo (1974). Phonological concreteness in natural generative grammar. In: Shuy R., Bailey C-J N (eds.) *Toward Tomorrow's Linguistics*. Washington D. C.: Georgetown University Press.**
- **Hooper, J.B. (1976). *An Introduction to Natural Generative Phonology*. New York: Academic Press.**

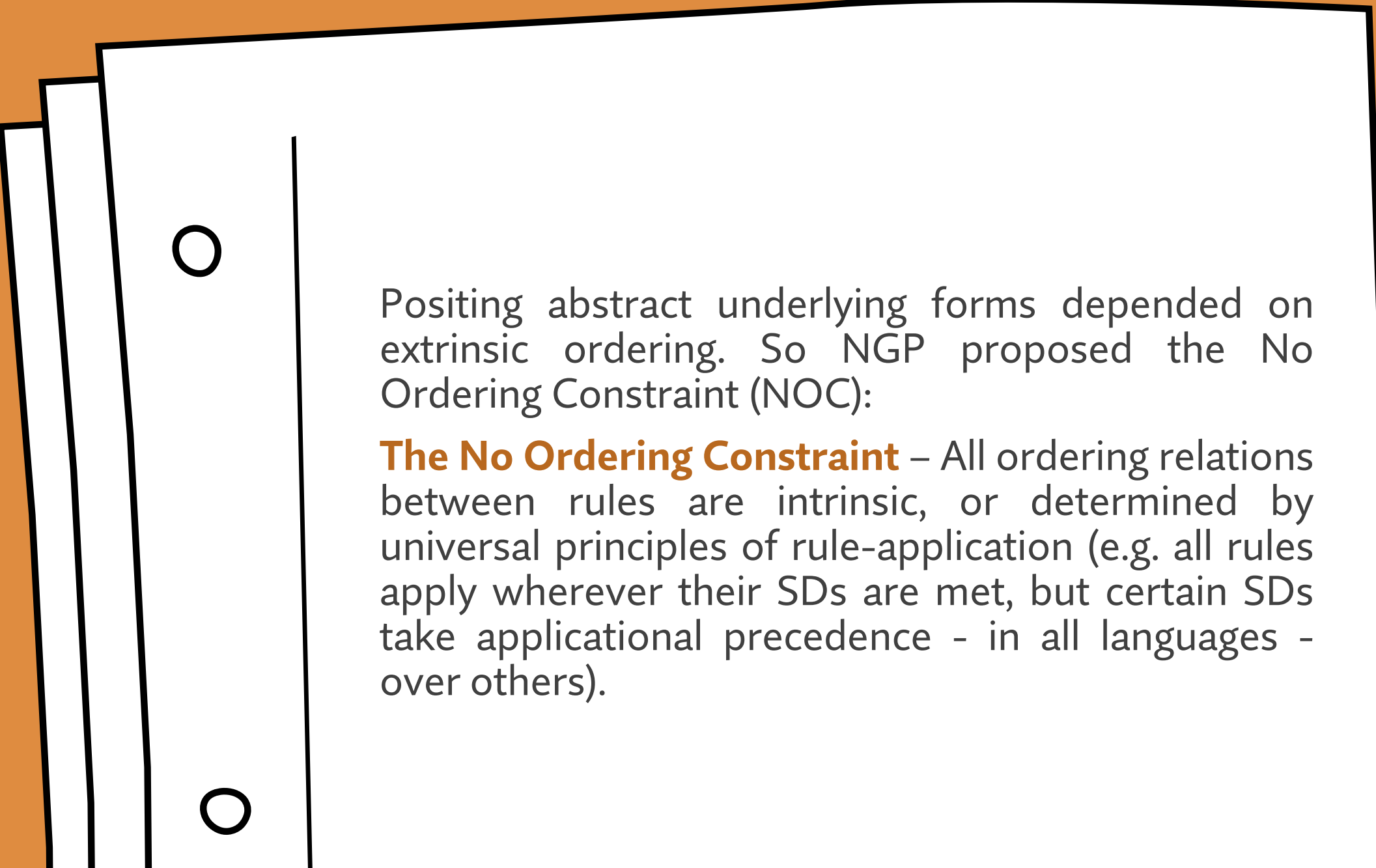
Natural Generative Phonology

- The issue of the SPE-school overusing abstractness in linking underlying forms with surface forms led to the emergence of alternate linear models of phonology.
- Natural Generative Phonology (NGP), emerged in the early 1970s from a bunch of papers by Theo Vennemann. Later, Hooper (1976), *An Introduction to Natural Generative Phonology* is one of these.
- NGP is a derivational approach, similar to Generative Phonology. As the title of the school suggests, it does not depart radically from mainstream generative phonology.



- The aim of NGP was to constrain abstract phonological representations.
- This was done by constraining rule ordering and underlying forms.





Positing abstract underlying forms depended on extrinsic ordering. So NGP proposed the No Ordering Constraint (NOC):

The No Ordering Constraint – All ordering relations between rules are intrinsic, or determined by universal principles of rule-application (e.g. all rules apply wherever their SDs are met, but certain SDs take applicational precedence - in all languages - over others).



Another condition proposed was the True Generalization Condition (TGC), to restrain underlying forms that could not be shown to correspond to their surface forms

The True Generalization Condition (TGC) - '... the rules speakers formulate are based directly on surface forms and ... relate one surface form to another, rather than relating underlying to surface form' (Hooper, 1976: 13). Under this condition all rules are 'surface-true', and no rule can refer to a 'nonexistent segment' ('existence' = (is equivalent to) 'surface occurrence').



On abstractness in NGP

- While Vennemann had proposed to rule out any underlying form that is not identical to a surface form, and prohibit rule ordering, Hooper (1976) assesses this proposal and concludes that it is too far fetched.

melody ['mɛlədi]	melodic [mə'lədɪk]
heretic ['hɛrɪtɪk]	heretical [hə'ɪtɪkəl]
demon ['dɪmən]	demonic [də'mɒnɪk]
telephone ['tɛləfəʊn]	telephonist [tə'ləfənist].

- A strict constraint on abstractness would mean that one of the surface forms would have to be posited as underlying.
- But if any of these roots were to be the underlying form, the vowel schwa must not have occurred in them as this vowel is derived by reduction from other vowels.
- Hooper suggests that we abandon underlying forms.

Hooper's alternative

- Hooper, while abandoning abstract underlying forms, revives the concept of archiphoneme of the type entertained by Trubetzkoy and the Prague School.
- Rules are now regarded as generalizations across surface forms rather than as the means of *generating* surface forms.
- She suggests that within NGP rules are representations are directly related to surface forms, phonological analysis is more concrete and more realistic than in SPE (Hooper, 1976).

“Speakers construct only generalizations that are surface true.”

Archiphoneme – A phonological unit characterized by the distinctive features which are common to two or more phonemes whose opposition is neutralized. E.g.: In English there is no opposition between [t] and [d] after an initial [s]. The unit written ‘t’ in ‘stop’ may be identified as an archiphoneme which is neither specifically voiceless nor specifically voiced, but has only the features that these consonants share.



- The formal apparatus of NGP offers no prospect of highly abstract underlying forms.
- NGP directs phonology back towards the concerns of phonemics. Hooper recognizes a distinction among rules that revives the traditional categorization into phonetic and morphophonemic rules.
- Hooper distinguishes between rules that refer only to phonetic information and reflect the 'automatic' pronunciation habits of a speaker, and rules that refer to grammatical or lexical contexts and often do admit exceptions



An example of NGP

Word final devoicing in German is a classic case of a neutralization process, and it is perfectly acceptable as a process in NGP since it expresses a true generalization --all word final obstruents are voiceless. Let's take a look at following examples:

A.	Ta[k]	"day"	B.	Ta[g]e	"days"	B → A
	lei[t]	"sorry"		lei[d]en	"to suffer"	
	Lo[p]	"praise"		lo[b]en	"to praise"	
	Rau[p]	"robbery"		Rau[b]en	"to rob"	
	We[k]	"way"		We[g]e	"ways"	
	Lan[t]	"land"		lan[d]en	"to land"	
	Grei[s]	"old man"		Grei[z]es	"old men"	
	Ra[t]	"advice"		ra[t]en	"to advise"	
	Vol[k]	"people"		Vol[k]e	"peoples"	
	Perisko[p]	"periscope"		Perisko[p]e	"periscopes"	
	Ho[f]	"courtyard"		Hö[f]e	"courtyards"	
	Ro[s]	"horse"		Ro[s]e	"horses"	

Both voiced and voiceless obstruents become voiceless when they occur at word final position.

Vennemann (1972)

spirant An older term for a *fricative. **Spirantization** is a historical process by which a stop consonant becomes a fricative.

Grimm's Law (the First sound shift) – sound law or series of sound laws by which consonants in Germanic differ systematically in manner of articulation from those of cognate forms in other Indo-European languages. In a traditional account, 'voiced aspirated' stops (e.g. d^h) change in Germanic to voiced ([d]); voiced stops (d) change in turn to voiceless ([t]); voiceless stops (t) change to fricatives ([θ])

Verner's Law The explanation by K. Verner of a set of exceptions to *Grimm's Law in which the Germanic reflex of a Proto-Indo-European voiceless stop was not, as predicted, a voiceless fricative. In words where it was, the accent in Proto-Indo-European, as attested by other languages such as *Vedic, had fallen on the preceding syllable: Gothic *brōpar* (-[θ]-), Vedic ^{bhrátar} - 'brother'. Where it was not, the accent had fallen elsewhere: Old English *fæder*, Vedic *pitár* 'father'.

Non uniqueness of phonological representation

In reading current discussions on the transcription of sounds by phonemes, one gets the impression of a tacit assumption that given the sounds of one language, there will be one and only one way of reducing them to a system of phonemes which represent the sound-system correctly. Since different writers do not in fact agree in the phonemic treatment of the same language, there arise then frequent controversies over the "correctness" or "incorrectness" in the use of phonemes. The main purpose of the present paper is to show that given the sounds of a language, there are usually more than one possible way of reducing them to a system of phonemes, and that these different systems or solutions are not simply correct or incorrect, but may be regarded only as being good or bad for various purposes.

(Source: Semantics Scholar)

Answer to phonetic neutralization – Archiphoneme (/p/ and /b/ after s)

Answer to phonemic neutralization – Markedness theory

Similarly, while German *bunt* [bunt] 'colorful' is /bunt/ and *Bund* [bunt] 'league' is /bund/ because the /t/ and /d/, which are neutralized into [t] in final position, occur phonetically in inflected forms, the [t] in uninflected words like *und* [unt] 'and' must be /t/, because according to this markedness system, voiceless obstruents are less highly marked than voiced obstruents.

and the velars. Should the segments underlying the homogeneous series [tɕ tɕ' ɕ] therefore be the mixed group /k k' ɕ/? This conflicts with the intuition of at least some native speakers who uniformly identify the palatals with the velars (Chao 1934: 48). Thus it seems that considerations of a higher order than lexical naturalness have to be invoked for the assignments of underlying representations in transformational-generative (TG) phonological theory, in this case, perhaps the differential uniformity and plausibility of the phonological processes required to actualize the posited representations phonetically.

A second example not immediately handled by Schane's proposal occurs in Hyman 1970. Hyman observes that in the Parisian dialect of French, positional neutralization of /en/, /in/, and /un/ into [ẽ] takes place, e.g., *brin* 'blade of grass' and *brun* 'brown (m.)' both are [brẽ]. *Brun* is /brun/ in Schane's system because of *brune* [bryn] 'brown (f.)'. But *brin* could be either /brɛn/ or /brin/ or /brun/. Schane's proposal rules out the first as more complex than the other two, but does not permit a decision between the last two because /i/ and /u/ are of equal complexity in Chomsky and Halle's markedness system (which may, of course, be a deficiency of this system). Again only principles of a higher order would allow us to select /brin/ over /brun/ in TG phonological theory, here perhaps the phonological distance between these two forms and the actual phonetic form, i.e. the number of rules involved and/or the number and content of features changed.

Vennemann gives one more example, this time from historical linguistics . Find it on page 107

Natural generative (NG) grammar (Vennemann 1971 a, b, c, d, e) is a generative theory of language which is characterized primarily by the absence of a provision for extrinsic rule ordering. The goal of this theory is to establish a system of universal constraints which is powerful enough to force the linguist analyzing a language to construct a grammar which directly models the mental representation of the generative and analytic capacity underlying that language. The primary sources of evidence for universal linguistic constraints are linguistic typology, linguistic change, and psycholinguistics (especially comparative language acquisition studies). The primary sources of language specific evidence are (besides the usual representative sample of discourses in their social context) the history of the language, especially its analogic tendencies as these are reflected in its most recent history and in neighboring dialects, and the body of psycholinguistic knowledge pertaining to that language.

Returning to the immediate problem of non-unique phonological representation, we see that the constraints contained in NG grammar for independent reasons automatically prevent this problem from ever arising in this theory. Given a morpheme with a segment x which is not subject to morphophonemic alternation, x is automatically fixed by the phonological rules of the language, with all its phonetic detail, regardless of whether or not the same segment is in a different morpheme derived from something else, e.g. from another segment, y . To give a concrete example, the fact that Faroese $[\text{ŋ}]$ occurs only before voiceless velar obstruents, and that $[\text{ŋ}]$ in the metathesized *svenkst* $[\text{sveŋkst}]$ (neuter nom. sg. derives from the radical $/\text{ŋ}/$ of */sveŋsk-/* 'Swedish', cf. *svenskur* $[\text{sveŋskʊr}]$ (masc. nom. sg.) by two phonological rules of the language, provides no justification for deriving from $/\text{ŋ}/$ the $[\text{ŋ}]$ of *banka* $[\text{beŋka}]$ 'knock', which does not alternate (except for palatalization). The lexical representation of the root of this word is $/\text{beŋk-}/$; at no level of representation can it be $/\text{benk-}/$, in this theory. Again, this does not mean that the nasal of this root is

Rule Types in NGP

1. Phonetically conditioned rules (P-rules)
2. Morphophonemic rules (MP-rules)
3. Syllable rules
4. Via rules

P-rules

- Phonetically conditioned rules (P-rules) are rules that are specified in purely phonetic terms, that is, phonological features, and phonological boundaries (syllable boundaries, pause boundaries). P-rules are automatic and have no exceptions (e.g., the rule aspirating syllable-initial voiceless stops in English). These rules correspond to the natural processes of natural phonology.
- “They are responsible for specifying the shape of the phonetic representation, in which some degree of variation is associated with every feature. Variation should be thought of as the extent to which a feature is altered and not as a matter of whether a rule applies or not. For instance, consider the process of flapping in English and what is sometimes thought of as a separate and highly variable process, flap-deletion. These should not be thought of as two separate processes, one which produces [D] and one that produces Ø. Rather they are one and the same process, a weakening of ambi-syllabic alveolar stops, which produces not just [D] and Ø, but dozens of articulations varying in a continuum from [D] to Ø. The extent to which the articulation is weakened, or the weakening is curtailed, depends on a variety of factors, such as the social situation, the tempo of speech, the degree of stress, and even certain lexical and grammatical factors.”

MP rules

- Morphophonemic rules (MP-rules) refer to morphological or syntactic categories, to arbitrary lexical categories, or to word or morpheme boundaries. Essentially, these include all morphosyntactically conditioned rules, producing 'phonological' (MP, not allophonic) alternations. The rule voicing fricatives in certain English plurals (wife, wives) is an example of an MP-rule, which contains both lexical and morphological information: it applies to only a small class of nouns (wife, house, but not safe, face), and it applies only in the plural.
- “In contrast with P-rules or processes, MP-rules make larger structural changes on the whole (e.g., the $k \rightarrow s$ of electric, electricity, versus the process $k \rightarrow ky$ of coo and key).”
- “MP-rule that has true exceptions. Consider the alleged S-voicing rule of English, which gives the alternations in con[s]erve and re[z]erve, con[s]ent, re[z]ent, [s]emblance, re[z]emble. Chomsky and Halle (1968) formulate a rule which voices /s/ after a V and a prefix boundary. This rule has dozens of surface exceptions, e.g., descent, recite, assemble, none of which, in even the most unguarded speech will ever undergo voicing to give *de[z]ent, *re[z]ite or *a[z]emble. Nor will the weakening of the boundary in re#sell, re#sew, re#sand ever yield a voiced fricative such as *re[z]ell, etc. The rule is unproductive, it is an MP-rule (if it is a rule at all), and it has true unyielding exceptions.”

Syllable rules

These specify the allowable shapes of syllables, and hence are conditions on morpheme and word structure.

Via rules

Great Vowel Shift A series of changes in late Middle English, by which close long vowels became diphthongs and other long vowels shifted one step closer. Thus, in the front series, [a:] > [ɛ:], > [e:], [e:] > [i:], [i:] > [a□]; in the back series, [ɔ:] > [o:], [o:] > [u:], [u:] > [a□]. Often interpreted as a unitary phenomenon; hence as a classic example of a *chain shift.

It is in consequence of these and other changes that [eI] in *name* (formerly [a:]) is spelled *a*, or [a□] in *shine* (formerly [i:]) spelled *i*. They are also the main factor in the development of vowel alternations between long [eI] and short [a] (in *sane/sanity*), long [a□] and short [I] (*divine/divinity*), and so on.

Via rules are used in NGP to relate pairs of words without deriving them from the same underlying form. Vennemann (1972) suggests that a correct account of the synchronic residue of the Great Vowel Shift of English, for example, is a via rule. Accordingly, ser[i:]ne and ser[ɛ]nity would not be derived from the same underlying form as proposed in SPE; rather the underlying forms would be essentially identical to the phonetic forms, and the lexical entries for both items would be marked to indicate that they are related to each other by a via rule:

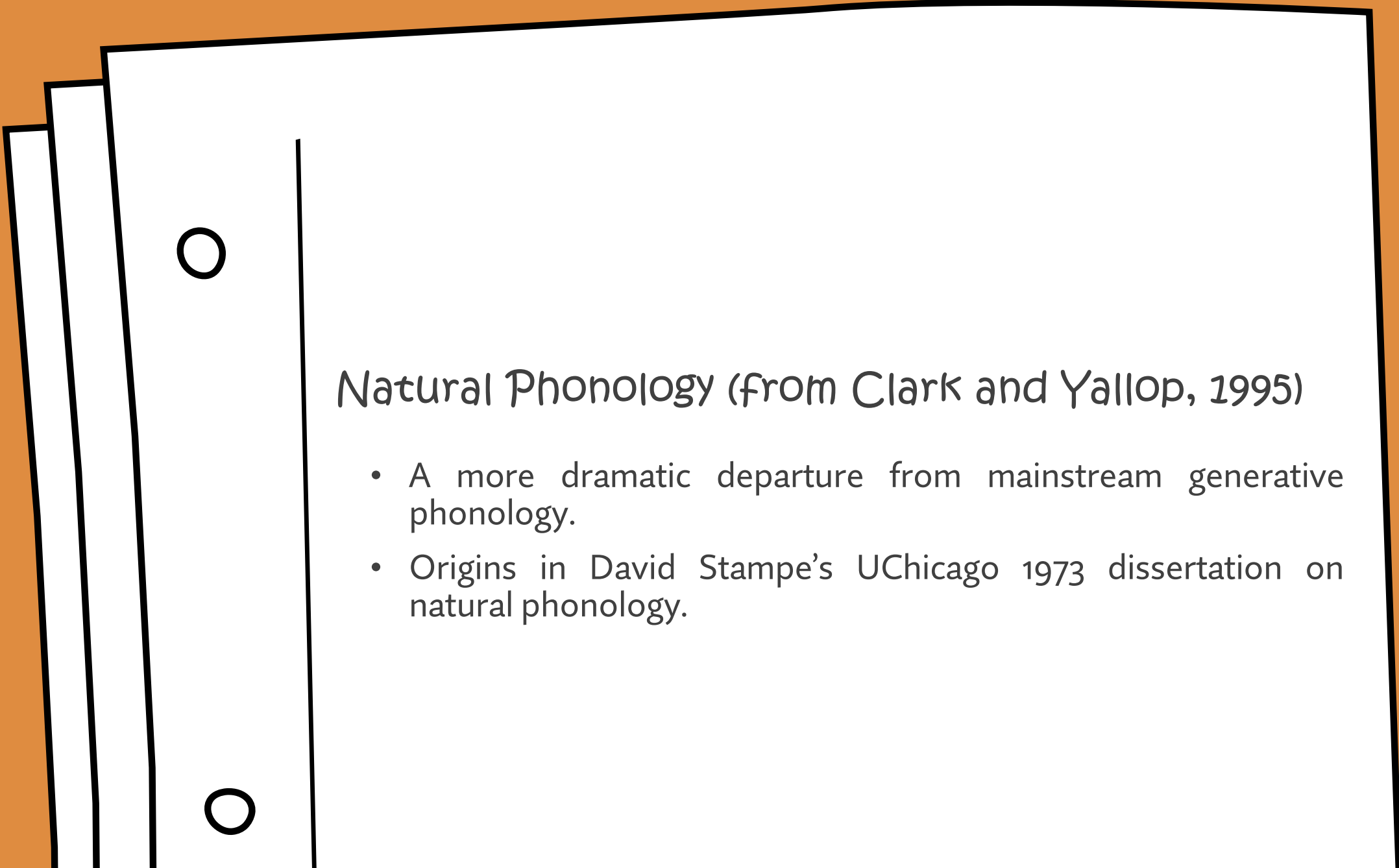
i: $\longleftrightarrow$ ɛ

Therefore, it would be possible for an individual speaker to perceive a relationship between serene and serenity, but to fail to see a relationship between obscene and obscenity. In this case, the former pair would be lexically marked as related by the via rule, whereas the latter pair would not.

Drawbacks of NGP

- It both overdifferentiates (by separating rules referring to boundaries from other P-rules and syllable-structure rules) and under-differentiates, by lumping all of the phonotactics under syllable structure. In the latter case, for instance, we lose sight of the possibility that syllable structure may be conditioned by word-position: e.g. in Kannada closed syllables are allowed, but only in non-final position, and all final syllables are open. This framework also takes no account of rules that operate in connected and casual speech.

- a. There were many cases, such as the difference between dialects on account of the differences in the pronunciation of forms, such as *writer/ rider*, that were argued to be inexplicable without the notion of extrinsic ordering in the grammar.
- b. The TGC would prevent postulating /*siŋg*/ for *sing*, as the UR form never actually occurs on the surface. The speakers of English, however, have no difficulty in assuming the psychological reality of the UR of *sing*, on account of a generalization common to forms such as *youn[g]er*, *han[g]er*, etc.
- c. Separating P-rules from MP rules would duplicate rules in the grammar, as argued in Halle (1962)



Natural Phonology (from Clark and Yallop, 1995)

- A more dramatic departure from mainstream generative phonology.
- Origins in David Stampe's UChicago 1973 dissertation on natural phonology.



- Those from the Natural Phonology school argue that the phonological component of language is not merely the result of CONVENTION, rather it is the way it is for very good reasons.
- It reflects, on the one hand, the auditory, articulatory and speech processing capacity of humans, and on the other hand the functions which language serves. They maintain that in all languages, phonemes and phonological processes are the residue of UNIVERSAL PHONOLOGICAL PROCESSES.
- Typically these are assimilation processes like homorganic nasal assimilation, palatalisation, nasalisation, voice assimilation, and so on.



Stampe on acquisition of phonology

- A phonological process is 'a mental operation that applies in speech to substitute for a class of sounds or sound sequences presenting a specific common difficulty to speech capacity of the individual, an alternative class identical but lacking the difficult property.' (Stampe, 1979, p.1)
- These processes are not rules of language, acquired as the child masters language, but reflections of what we might call the child's inbuilt tendencies. Thus., by the very nature of the human articulatory and perceptual organism, a child will prefer to articulate plosives as voiceless rather than voiced (because of the relative difficulty of maintaining voicing while the supraglottal tract is closed off), or will prefer to nasalize vowels next to nasal consonants (again for articulatory reasons)

The application of phonological processes is not as straightforward as simple examples might suggest. Not only are there many such processes, but some of them are contrary to others. For example, a process of vowel denasalization reflects the goals of articulatory ease and auditory distinctness in vowel production; but this is to some extent countered by the process of nasalizing vowels next to nasal consonants (Stampe 1979, pp. 17–23). In early stages of language acquisition, the unconstrained operation of natural processes will tend to reduce every potential utterance to something like a monosyllabic [pa] (Stampe 1979, pp. xvii, 2). As the child comes closer to an adult competence, processes will be suppressed or limited in response to the demands of the phonological system. Hence, if acquiring a language in which nasalized vowels are distinctive, a child will have to suppress the relevant natural processes and thus achieve control of vowel nasalization; but in a language in which vowel nasalization is not distinctive, the natural process of nasalizing vowels next to nasal consonants may persist as an ‘allophonic rule’ of adult speech (Stampe 1979, pp. 27–8).

Stampe appears to turn generative phonology on its head. What we thought of as rules constituting a phonological system are now seen as processes motivated by the nature of production and perception. Phonological acquisition is a matter of suppressing or constraining innate tendencies rather than of learning rules. Stampe does leave room for phonological rules, however. These are indeed acquired, but they differ sharply from processes. The English alternation of /g/ and /dʒ/ (as in, for example, *analogous* with [g] but *analogy* with [dʒ]) is governed by an acquired rule. Unlike a natural process, such a rule is open to exceptions and easily suspended: many speakers fail to apply the rule consistently and pronounce *analogous* with [dʒ] or *pedagogy* with [g], for example, and even those who follow the rule could easily produce the 'wrong' pronunciations if they wished. By contrast, the phonetic consequences of natural processes, such as aspiration of voiceless plosives and lengthening of vowels before voiced obstruents, are much harder for native speakers to discern and overcome (Stampe 1979, pp. 45–7).

Among the various North American 'schools' of phonology which represent reactions to orthodox generative phonology, natural phonology is the least inclined to proclaim its faithfulness to generative principles. In their useful outline of natural phonology, Donegan and Stampe (1979) appeal to phonological traditions that are much older than SPE: 'Natural phonology is a modern development of the oldest explanatory theory of phonology . . . Its basic thesis is that the living sound patterns of languages, in their development in each individual as well as in their evolution over the centuries, are governed by forces implicit in human vocalization and perception' (1979, p. 126).

Donegan and Stampe claim that their theory is natural because it seeks to explain why language is the way it is. The theory offers genuine explanation by presenting language not as merely conventional but as a 'natural reflection' of the needs, capacities, and world of its users' (Donegan and Stampe 1979, p. 127). Donegan and Stampe are critical of positivism, which gives priority to exhaustive scientific description (p. 127; cf. section 11.1 above). They reject (underlying) morphophonemic representation in favour of a more traditional phonemic representation (which, following Sapir, they interpret as reflecting the phonological intention of speech; pp. 158ff. and especially 163–7). And they conclude that although both structuralist and generative phonology have a well-developed methodology, neither of them is a theory in the true sense, since neither is genuinely open to falsification by data (pp. 167–8).